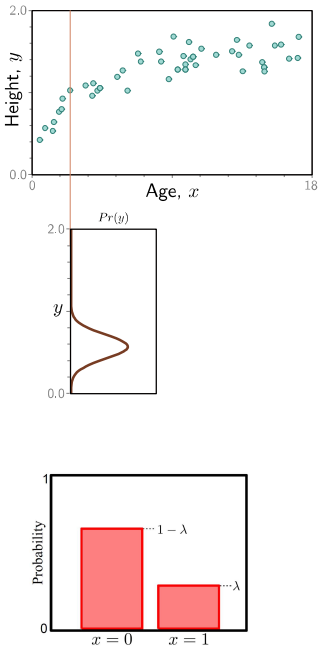
Last Time

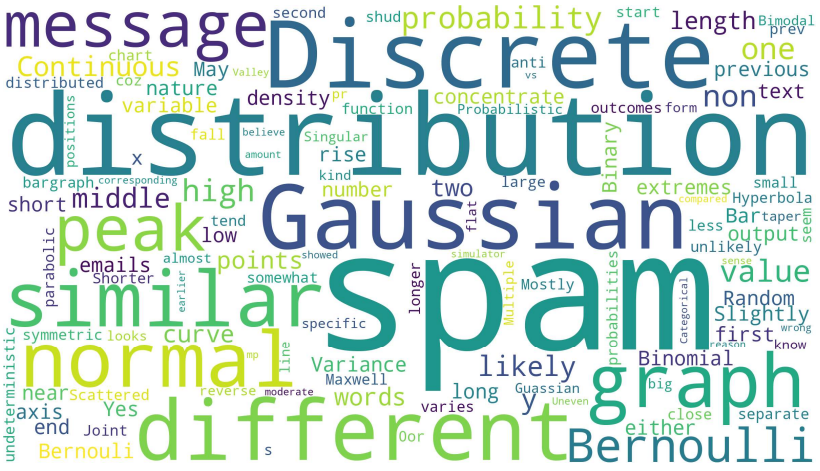
- Some distributions used over and over again
- Therefore, useful to know the concept
  - If we know the mean and variance, we can understand these better
  - However, do not memorize the pmf (pdf), means and variance, we can always look them up



2

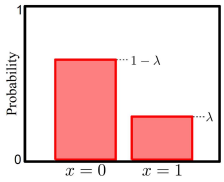
Engages with Question

What is the nature of the output distribution for text classification?

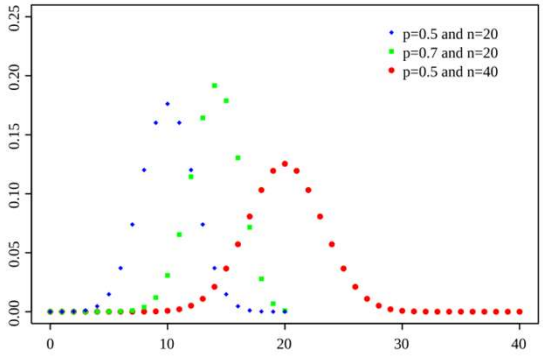


3

Agenda

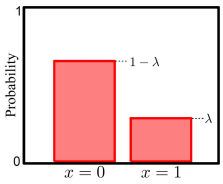


- Last time
  - Some distributions used over and over again
    - If we know the mean and variance, we can understand these better
  - Bernoulli
- Today
  - Binomial
  - Geometric
  - Hypergeometric
  - Moment generation functions

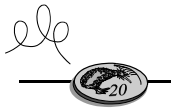


5

Bernoulli



- Some distributions used over and over again
- Last time
  - W ~ Bern(λ)
    - Expectation: λ, Variance: $\lambda(1-\lambda)$
- Physical analogy
  - Biased coin



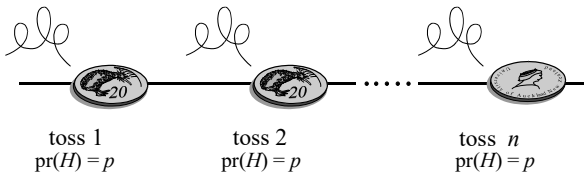
toss 1
pr(H) = p

6

Binomial

- Physical model for situations of a series of trials
  - Each trial has only two outcomes: success and failure;
  - p = pr(success) is the same for every trial; and
  - trials are independent
- Expectation and Variance?

Biased coin model



toss 1 pr(H) = p | toss 2 pr(H) = p | toss n pr(H) = p

Perform n tosses and count X = # heads

$$P(X = i) = \binom{n}{i} p^i (1-p)^{n-i}$$

7

Example: US Presidents

- The Economist magazine reported that US presidents had 61 daughters and ~90 sons
- Expected value is approximately 75
- What is the probability that there are 89 or more boys?
  - (1 - binocdf (89, 150, 0.5)) in octave
- Probability theory calculates that the probability of having 89 or more sons in 151 'experiments' is 0.008799 (~0.8%), a very small number

octave convention
use 88

Were US presidents special?

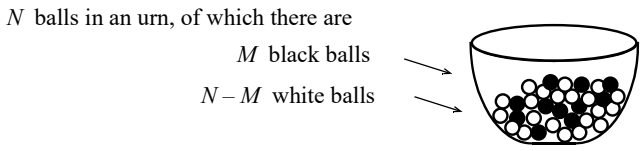
Overview of Presidential Children

- **Total Recorded (Historical baseline up to Clinton):** 89 boys and 61 girls (150 total)

8

Hypergeometric

- Urn model is actually "hypergeometric" distribution
- We chose Y balls *without replacement* and ask how many are black
- However, the binomial approximation to the urn Model Approximation is very useful
  - Y items sampled, count X as the number of people which have a characteristic
  - Approximation: X ~ Bin(Y, M/N)



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Example: Doctors and Exercise

- A 2018 study followed a nationwide cohort of Norwegian doctors. In its 2014 survey, 307 of 526 respondents reported that they asked patients about exercise habits
  - You are choosing a doctor for your pre-diabetic uncle
  - You have a choice of three doctors in your clinic
  - What is the probability that none of the three doctors will ask about exercise?
- This problem fails the binomial requirement of "sampling with replacement" and independence of opinion
  - But we will use the binomial approximation since 3/526 is very small.
  - P(X<=0) = binocdf(0, 3, q=219/526) = 19.8%

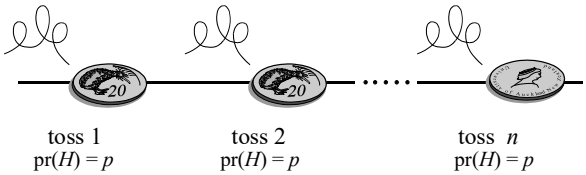
Is the clinic a good one?

10

Answers a different question than what is posed (as in class lecture probably)

Use $\frac{307}{526}$

Binomial distribution



Perform n tosses and count X = # heads

- Expectation
- Variance

$$P(X = i) = \binom{n}{i} p^i (1-p)^{n-i}$$

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Binomial Distribution

- Defining indicator variables, a binomial random variable *X* is the *sum* of indicator random

$$P(X = i) = \binom{n}{i} p^i (1-p)^{n-i}$$

$$X_i = \begin{cases} 1, & \text{trial } i \text{ succeeds (probability } p) \\ 0, & \text{otherwise} \end{cases}$$

$$E(X) = E\left(\sum_{i=1}^{n} X_i\right) = np$$

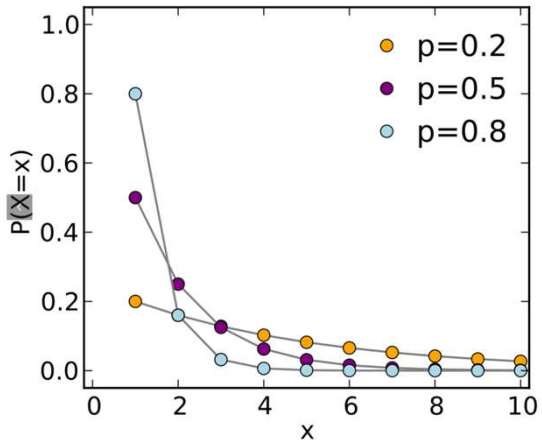
$$\mathrm{Var}(X) = \mathrm{Var}\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} \mathrm{Var}(X_i) = np(1-p)$$

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Geometric Distribution

$$P_X(k) = \begin{cases} p(1-p)^{k-1} & \text{for } k = 1, 2, 3, \ldots \\ 0 & \text{otherwise} \end{cases}$$

- X ~ Geom(p)
- A sequence of independent Bernoulli trials until you observe the first success
- "Memoryless"
- E[X] = ?
- E[X] = 1.p + (1-p)[1 +E[X]]
- Var(X) = $(1-p)/p^2$



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